$$E^{\circ}_{cell} = \left(E^{\circ}_{ox}\right) + \left(E^{\circ}_{red}\right)$$
$$= -1.09 + 0.771$$
$$= -0.319V$$

E°_{cell} is – ve; ΔG is +ve and the cell reaction is non spontaneous. Hence Fe^{3+} cannot oxidises Br^{-} to Br_2 .

15. $\left(E^{\circ}_{ox}\right)_{Fe|Fe^{2+}} = 0.44V$ and $\left(E^{\circ}_{red}\right)_{Cu^{2+}|Cu} = 0.34V$.

These +ve emf values shows that iron will oxidise and copper will get reduced i.e., the vessel will dissolve. Hence it is not possible to store copper sulphate in an iron vessel.

16. Metals having higher oxidation potential will liberate H_2 from H_2SO_4 . Hence, the metal M_1 having + xV, oxidation potential will liberate H_2 from H_2SO_4 .
17. oxidation potential of M_1 is more +ve than the oxidation potential of Fe which indicates that it will prevent iron from rusting
18. Cell reactions:

Oxidation at anode: $Cd\,(s) \rightarrow Cd^{2+}(aq) + 2e^{-}$ $\qquad \left(E^{\circ}_{ox}\right)_{Cd|Cd^{2+}} = 0.4V$

Reduction at cathode: $Cu^{2+}(aq) + 2e^{-} \rightarrow Cu\,(s)$ $\qquad \left(E^{\circ}_{red}\right)_{Cu^{2+}|Cu} = 0.34V$

$$Cd(s) + 2e^{-} \rightarrow Cd^{2+}(aq) + Cu(s)$$
$$E^{\circ}_{cell} = \left(E^{\circ}_{ox}\right) + \left(E^{\circ}_{red}\right)_{cathode}$$
$$= 0.4 + 0.34$$
$$= 0.74V.$$

emf is +ve, so ΔG is (-)ve, the reaction is feasible.

19. Oxidation at anode:

$2H_2(g) + 4OH^{-}(aq) \rightarrow 4H_2O\,(l) + 4e^{-}$

1 mole of hydrogen gas produces 2 moles of electrons at 25^0C and 1 atm pressure, 1 mole of hydrogen gas occupies = 22.4 litres

$$\therefore \text{ no. of moles of hydrogen gas produced} = \frac{1 \text{ mole}}{22.4 \text{ litres}} \times 44.8 \text{ litres}$$
$$= 2 \text{ moles of hydrogen}$$

$\therefore$ 2 of moles of hydrogen produces 4 moles of electron i.e., 4F charge.

We know that $Q = It$

$$I = \frac{Q}{t}$$
$$= \frac{4F}{10 \text{ mins}}$$
$$= \frac{4 \times 96500 \text{ C}}{10 \times 60 \text{ s}}$$
$$I = 643.33 \text{ A}$$

Electro deposition of copper

$Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)$

2F charge is required to deposit

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